Project Documentation

Auto Ticket Classification using Flow Designer

Team ID: SWTID-2026-3564
Date: 29.9.2026

1. Introduction

- **Project Title:** Auto Ticket Classification using Flow Designer
- **Team Members:** Team ID SWTID-2026-3564

Team Member	Role
[Member 1 name]	Team Lead - flow design and configuration
[Member 2 name]	Instance setup, groups and categories
[Member 3 name]	Testing and validation
[Member 4 name]	Documentation and reporting

2. Project Overview

- **Purpose:** To automatically classify incoming support tickets and assign them to the correct support group using a flow built in ServiceNow Flow Designer, so that manual triage is removed and tickets are resolved faster.
- **Features:**
- Automatic trigger when a new incident is created.
- Keyword-based classification of the ticket description into categories (Network, Software, Hardware, Access).
- Automatic assignment of the ticket to the matching assignment group.
- Default group for tickets that match no rule.
- Low-code visual design that is easy to modify and extend.
- Execution history in Flow Designer for monitoring and debugging.

3. Architecture

This project is built on the ServiceNow platform. The three layers of the standard documentation format map to ServiceNow components as follows.

- **Frontend:** Users raise tickets through the ServiceNow Service Portal or the standard Incident form. Support agents view classified tickets in the Incident list and workspace. No custom frontend code is needed.
- **Backend:** The logic runs in Flow Designer. The flow has a trigger (Incident - Created), condition blocks that check keywords in the short description, and Update Record actions that set the Category and Assignment Group.
- **Database:** Data is stored in the ServiceNow Incident table [incident]. The fields used are given below.

Field	Type	Use
number	String	Ticket number
short_description	String	Text used for classification
category	Choice	Set by the flow
assignment_group	Reference (sys_user_group)	Set by the flow
priority	Choice	Priority of the ticket
caller_id	Reference (sys_user)	User who raised the ticket

4. Setup Instructions

- **Prerequisites:** A ServiceNow Personal Developer Instance (from developer.servicenow.com), a user with the admin role (or flow_designer and itil roles), and a web browser.
- **Installation:**

1 Sign in to developer.servicenow.com and request a Personal Developer Instance.

2 Log in to the instance as admin.

3 Create the required assignment groups (for example Network Team, Software Team, Hardware Team, Service Desk) under User Administration > Groups.

4 Confirm the Category choices on the Incident table.

5 Open All > Process Automation > Flow Designer and create a new flow named Auto Ticket Classification.

6 Set the trigger to Created on the Incident table, add the condition blocks and Update Record actions, then Save and Activate.

5. Folder Structure

ServiceNow does not use project folders. The equivalent organisation of the components is shown below.

- **Client:** Service Portal / Incident form (user-facing side) where tickets are raised. This uses the standard ServiceNow interface.
- **Server:** Flow Designer objects that run on the instance.

```
Auto Ticket Classification
|-- Flow Designer
|   +-- Flow: Auto Ticket Classification
|       |-- Trigger: Incident - Created
|       |-- If: Network keywords -> Update Record
|       |-- If: Software keywords -> Update Record
|       |-- If: Hardware keywords -> Update Record
|       |-- If: Access keywords -> Update Record
|       +-- Else: Default group
|-- User Administration
|   +-- Groups (assignment groups)
```

```
+-- Tables
    +-- incident
```

6. Running the Application

The flow runs on the ServiceNow instance, so no local servers are started. Use the steps below instead of npm start.

- **Frontend:** Open the Incident form (All > Incident > Create New) or the Service Portal and submit a ticket.
- **Backend:** Activate the flow in Flow Designer. It then runs automatically on every new incident. Check Flow Designer > Executions to see each run.

7. API Documentation

The flow itself has no custom endpoints. Tickets can also be created and checked through the standard ServiceNow Table API, and the flow classifies them in the same way.

Method	Endpoint	Description
POST	/api/now/table/incident	Create a new incident (triggers the flow)
GET	/api/now/table/incident/{sys_id}	Get an incident and view its category and group
GET	/api/now/table/incident?sysparm_query=category=network	List incidents by category
PATCH	/api/now/table/incident/{sys_id}	Update an incident

Example request (POST)

```
{ "short_description": "VPN is not connecting", "caller_id": "user_sys_id" }
```

Example response (after the flow runs)

```
{ "result": {
    "number": "INC0010001",
    "short_description": "VPN is not connecting",
    "category": "network",
    "assignment_group": { "value": "group_sys_id" }
} }
```

8. Authentication

- Users log in to the ServiceNow instance with a user name and password. Sessions are managed by the platform.
- API calls use Basic Authentication or OAuth 2.0 tokens.
- Authorization is role-based. Access control rules (ACLs) and roles such as admin, itil and flow_designer decide who can create, view, edit and run flows and incidents.

9. User Interface

The interface is the standard ServiceNow UI. The screens below show the main features.

[Screenshot: Incident form / Service Portal]

[Screenshot: Flow Designer canvas]

[Screenshot: Incident list with auto-assigned category and group]

10. Testing

- **Strategy:** Functional testing with sample incidents for every category plus one non-matching ticket, checking the category and assignment group set by the flow.
- **Tools:** ServiceNow Incident form and Flow Designer Executions view.

Test Case	Sample Short Description	Expected Result	Status
TC-01	VPN is not connecting	Network, Network Team	Pass
TC-02	Unable to install application	Software, Software Team	Pass
TC-03	Laptop keyboard not working	Hardware, Hardware Team	Pass
TC-04	Forgot my password	Access, Service Desk	Pass
TC-05	General enquiry	Default group	Pass

11. Screenshots or Demo

Screenshots of the working flow and classified tickets are given below. A demo link can be added here.

[Screenshot: Flow execution details]

[Screenshot: Final classified incident]

Demo Link: [Add link here]

12. Known Issues

- Keyword matching can miss tickets that use unusual wording or spelling mistakes.
- A ticket that contains keywords from two categories is placed in the first matching category.
- The keyword list must be updated by hand when new categories are added.

13. Future Enhancements

- Use ServiceNow Predictive Intelligence (machine learning) for smarter classification.
- Predict priority and set SLA automatically.
- Send email or chat notifications to users and assignees.
- Support tickets written in multiple languages.
- Extend the flow to requests, problems and change tickets.